

1. NAME OF THE MEDICINAL PRODUCT

TRAVATAN

40 micrograms/ml eye drops, solution

2. QUALITATIVE AND QUANTITATIVE COMPOSITION

Travoprost 40 µg /mL, polyquaternium-1, polyoxyethylene hydrogenated castor oil 40(HCO-40), boric acid, mannitol, sodium chloride, propylene glycol, sodium hydroxide and/or hydrochloric acid (to adjust pH); purified water.

3. PHARMACEUTICAL FORM

Eye drops, solution.

Clear, colourless solution.

4. CLINICAL PARTICULARS

4.1 Therapeutic indications

Decrease of elevated intraocular pressure in patients with ocular hypertension or open-angle glaucoma (see section 5.1).

4.2 Posology and method of administration

Use in adults including the elderly population

1 drop of TRAVATAN eye drops in the conjunctival sac of the affected eye(s) once daily.

Optimal effect is obtained if the dose is administered in the evening.

Paediatric population

The efficacy and safety of TRAVATAN® eye drops in patients below the age of 18 years have not been established and its use is not recommended in these patients until further data become available.

Hepatic and renal impairment

TRAVATAN eye drops has been studied in patients with mild to severe hepatic impairment and in patients with mild to severe renal impairment (creatinine clearance as low as 14 ml/min). No dosage adjustment is necessary in these patients.

Method of Administration

TRAVATAN eye drop is for ocular use.

Nasolacrimal occlusion or gently closing the eyelid after administration is recommended. This may reduce the systemic absorption of medicinal products administered via the ocular route and result in a decrease in systemic adverse reactions.

If a dose is missed, treatment should be continued with the next dose as planned. The dose should not exceed one drop in the affected eye(s) daily.

When substituting another ophthalmic anti-glaucoma agent with TRAVATAN eye drops, the other agent should be discontinued and TRAVATAN eye drops should be started the following day.

If more than 1 topical ophthalmic medicinal product is being used, the medicinal products must be administered at least 5 minutes apart.

The patient should remove the protective overwrap immediately prior to initial use.

To prevent contamination of the dropper tip and solution, care must be taken not to touch the eyelids, surrounding areas or other surfaces with the dropper tip of the bottle.

4.3 Contraindications

Hypersensitivity to the active substance or to any of the excipients.

4.4 Special warnings and precautions for use

Eye Colour Changes

TRAVATAN eye drops may gradually change the eye colour by increasing the number of melanosomes (pigment granules) in melanocytes. Before treatment is instituted, patients must be informed of the possibility of a permanent change in eye colour. The change in iris colour occurs slowly and may not be noticeable for months to years.

Periorbital and eye lid changes

Periorbital and/or eyelid skin darkening has been reported in association with the use of TRAVATAN eye drops.

Periorbital and lid changes including deepening of the eyelid sulcus have been observed with prostaglandin analogues.

TRAVATAN eye drops may gradually change eyelashes in the treated eye(s); these changes were observed in about half of the patients in clinical trials and include: increased length, thickness, pigmentation, and/or number of lashes.

Aphakic Patients

Macular oedema has been reported during treatment with prostaglandin F2a analogues. Use TRAVATAN eye drops with caution in aphakic patients, pseudophakic patients with torn posterior lens capsule or anterior chamber lenses, or in patients with known risk factors for cystoid macular oedema.

Iritis/Uveitis

TRAVATAN eye drops should be used with caution in patients with active intraocular inflammation, as well as patients with predisposing risk factors for uveitis.

4.5 Interaction with other medicinal products and other forms of interaction

No clinical relevant interactions have been described.

4.6 Fertility, pregnancy and lactation

Fertility

There are no data on the effects of TRAVATAN eye drops on human fertility. Animal studies showed no effect of travoprost on fertility at doses more than 250 times the maximum recommended human ocular dose.

Pregnancy

There are no or limited amount of data from the use of TRAVATAN eye drops in pregnant women. Studies in animals with travoprost have shown reproductive toxicity. TRAVATAN eye drops should not be used during pregnancy unless clearly necessary.

Breastfeeding

It is unknown whether topical travoprost/metabolites are excreted in human milk. Animal studies have shown excretion of travoprost and metabolites in breast milk. The use of TRAVATAN eye drops by breastfeeding mothers is not recommended.

4.7 Effects on ability to drive and use machines

Temporary blurred vision or other visual disturbances may affect the ability to drive or use machines. If blurred vision occurs at instillation, the patient must wait until the vision clears before driving or using machinery.

4.8 Undesirable effects

The following adverse reactions have been reported during clinical studies with TRAVATAN eye drops and are classified according to the subsequent convention: very common ($\geq 1/10$), common ($> 1/100$ to $< 1/10$), uncommon ($> 1/1,000$ to $\leq 1/100$), rare ($> 1/10,000$ to $\leq 1/1000$), or very rare ($\leq 1/10,000$). Within each frequency grouping, adverse reactions are presented in order of decreasing seriousness.

System Organ Classification	MedDRA Preferred Term (v.18.0)
Immune system disorders	<i>Uncommon:</i> hypersensitivity
Nervous system disorders	<i>Uncommon:</i> headache <i>Rare:</i> dizziness, dysgeusia
Eye disorders	<i>Very common:</i> ocular hyperaemia <i>Common:</i> eye pain, eye pruritus, dry eye, eye irritation, iris hyperpigmentation, ocular discomfort <i>Uncommon:</i> corneal erosion, punctate keratitis, keratitis, iritis, cataract, visual acuity reduced, conjunctivitis, anterior chamber inflammation, blepharitis, vision blurred, photophobia, periorbital oedema, eyelids pruritus, eye discharge, eyelid margin crusting, lacrimation increased, erythema of eyelid, growth of eyelashes <i>Rare:</i> uveitis, iridocyclitis, ophthalmic herpes simplex, conjunctival follicles, conjunctival oedema, hypoaesthesia eye, eye inflammation, trichiasis, anterior chamber pigmentation, asthenopia, eye allergy, eczema eyelids, eyelid irritation, eyelash hyperpigmentation, eyelash thickening
Cardiac disorders	<i>Rare:</i> heart rate decreased, palpitations
Vascular disorders	<i>Rare:</i> hypertension, hypotension
Respiratory, thoracic and mediastinal disorders	<i>Rare:</i> asthma, dyspnoea, dysphonia, cough, rhinitis allergic, oropharyngeal pain, nasal discomfort, nasal dryness.
Gastrointestinal disorders	<i>Rare:</i> dry mouth, constipation
Skin and subcutaneous tissue disorders	<i>Uncommon:</i> skin hyperpigmentation, hypertrichosis <i>Rare:</i> skin discolouration, madarosis, erythema, hair colour changes, rash
Musculoskeletal and connective tissue disorders	<i>Rare:</i> arthralgia, musculoskeletal pain
General disorders and administration site conditions	<i>Rare:</i> asthenia

Additional adverse reactions identified from post-marketing surveillance with TRAVATAN eye drops include the following. Frequencies cannot be estimated from the available data. Within each System Organ Class adverse reactions are presented in order of decreasing seriousness.

System Organ Classification	MedDRA Preferred Term (v.18.0)
Psychiatric disorders	Depression, anxiety, insomnia
Eye disorders	Macular oedema, lid sulcus deepened
Ear and labyrinth disorders	Tinnitus
Cardiac disorders	Arrhythmia, tachycardia, chest pain
Respiratory, thoracic and mediastinal disorders	Epistaxis
Gastrointestinal disorders	Diarrhoea, vomiting, nausea, abdominal pain
Skin and subcutaneous tissue disorders	Pruritus
Renal and urinary disorders	Dysuria, urinary incontinence
Investigations	Prostatic specific antigen increased

4.9 Overdose

A topical overdose is not likely to occur or to be associated with toxicity.

Treatment of a suspected oral ingestion is symptomatic and supportive.

5. PHARMACOLOGICAL PROPERTIES

5.1 Pharmacodynamic properties

Pharmacotherapeutic Group: Ophthalmologicals-antiglaucoma preparations and miotics-prostaglandin analogues

ATC code: S01E E04

Mechanism of action

Travoprost, a prostaglandin F2 α analogue, is a highly selective full agonist which has a high affinity for the prostaglandin FP receptor, and is believed to reduce IOP by increasing the outflow of aqueous humour via trabecular meshwork and uveoscleral pathways. Reduction of IOP in human starts about 2 hours after administration and maximum IOP reduction is reached within 12 hours. Significant lowering of IOP can be maintained for periods exceeding 24 hours with a single dose.

As primary therapy, TRAVATAN eye drops, dosed once daily, reduced IOP 7 to 9 mmHg. Stable diurnal IOP reductions were achieved as early as 2 weeks after initiation of therapy and were maintained over 6 to 12 month treatment periods in 3 well-controlled studies.

Pharmacodynamic effects

In addition to reducing IOP, travoprost has been shown to increase optic nerve head blood flow, and decrease tear film stability and tear secretion. Travoprost does not affect respiration rate/volume or systolic blood pressure during exercise and recovery. Prostaglandin F2 α analogs can induce the anagen phase in hair follicles and stimulate melanogenesis in the skin.

Travoprost 0.004% eye drops preserved with polyquaternium-1 induced minimal ocular surface toxicity, compared to eye drops preserved with benzalkonium chloride, on cultured human corneal cells and following topical ocular administration in rabbits.

Pharmacodynamic/Pharmacokinetic relationship

See Section 5.2

Data from clinical trials

In a clinical trial, patients with open-angle glaucoma or ocular hypertension who were treated with Travoprost 0.004% eye drops (polyquaternium-preserved) dosed QD in the evening demonstrated 8-9 mmHg reductions (approximately 33%) in IOP from a baseline range of 24-36 mmHg. Data on adjunctive administration of Travoprost 0.004% eye drops with timolol 0.5% and limited data with brimonidine 0.2% collected during clinical trials showed an additive effect of Travoprost 0.004% eye drops with these concomitant medications. No clinical data are available on adjunctive use with other ocular hypotensive medications. Travoprost 0.004% eye drops is generally well-tolerated and safe. The most common side effect is hyperemia, as observed with other ophthalmic prostaglandin analogs.

Pediatric population

See Section 4.2.

5.2 Pharmacokinetic properties

Absorption

Travoprost is an isopropyl ester prodrug. It is absorbed through the cornea where the isopropyl ester is hydrolysed to the active free acid. Studies in rabbits have shown maximum concentrations of approximately 20 ng/g of travoprost free acid in aqueous humour were achieved within 1-2 hours of topical ocular dosing. Aqueous humour concentrations of travoprost free acid declined with a half-life of approximately 1.5 hours. Low concentrations of travoprost free acid are also found in plasma following topical dosing.

Distribution

Following topical ocular administration to humans, low systemic exposure to active free acid was observed, with peak plasma concentrations of 20 pg/ml or less observed between 10 and 20 minutes post-dose. Plasma concentrations declined rapidly to below the 10 pg/ml assay quantitation limit within 1 hour of administration. Trace plasma concentrations of travoprost may be present immediately following dosing in some subjects.

Biotransformation

Metabolism is the major route of clearance for both travoprost and its free acid in nonclinical species. The systemic metabolic pathways parallel those of endogenous prostaglandin F_{2α} which are characterised by reduction of the 13-14 double bond, oxidation of the 15-hydroxyl to a ketone and β-oxidative cleavages of the carboxylic acid side chain.

Elimination

Following administration of radiolabelled travoprost to rats, approximately 95% of the dose was eliminated within 24 hours. Approximately, 75% of the dose was eliminated in the feces and the remainder was excreted in urine.

Linear/non-linear pharmacokinetics

Travoprost exhibits linear pharmacokinetics in both ocular tissues and plasma after topical ocular administration.

Pharmacokinetic/pharmacodynamic relationship(s)

Pharmacokinetic and pharmacodynamics relationship has not been established for travoprost after topical ocular administration.

Pharmacokinetics in special populations

The systemic pharmacokinetics of Travoprost 0.004% eye drops has been studied in patients with mild to severe hepatic impairment as well as in patients with mild to severe renal impairment (creatinine clearance as low as 14 mL/minute). No dose adjustment is required in these populations.

Pediatric pharmacokinetics

The systemic pharmacokinetics of travoprost after topical ocular administration in patients between the ages of 2 months to less than 18 years of age showed a similar concentration range to that of adults, with most plasma samples below the limit of quantitation of 10 pg/mL.

5.3 Preclinical Safety Data

Non-clinical data reveal no special hazard for humans based on conventional studies of safety pharmacology, repeated-dose toxicity, genotoxicity, and carcinogenic potential. Adverse reproductive and developmental toxicity was observed in animals at exposure levels of travoprost similar to clinical exposure levels and is possibly relevant to clinical use.

6. PHARMACEUTICAL PARTICULARS

6.1 List of excipients

Polyquaternium-1

Polyoxyethylene hydrogenated castor oil 40 (HCO-40)

Boric acid (E284)

Mannitol (E421)

Sodium chloride

Propylene glycol (E1520)

Sodium hydroxide and/or hydrochloric acid (to adjust pH)

Purified water

6.2 Incompatibilities

None known.

Specific in vitro interaction studies were performed with TRAVATAN eye drops and medicinal products containing thiomersal. No evidence of precipitation was observed.

6.3 Storage Condition

Do not store above 30°C.

6.4 Nature and contents of container

4 ml oval bottle with dispensing plug and screw cap, all plastic, presented in an overwrap. Each 4 mL bottle will contain 2.5 mL of solution.

Cartons containing 1 bottle.

7.0 Product Registration Holder

Novartis Corporation (Malaysia) Sdn. Bhd.
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No. 8, Jalan SS21/37, Damansara Uptown,
47400 Petaling Jaya, Selangor.

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